Yinan Huang

yhuang903@gatech.edu | https://yinanhuang.github.io/| Google Scholar

Education

Georgia Institute of Technology, GA, USA — Sep 2023 - 2027 (Expected)
Ph.D. in Machine Learning

Duke University, NC, USA — Sep 2020 - May 2023
M.S. in Electrical and Computer Engineering

Sun Yat-sen University, Guangdong, China — Sep 2016 - May 2020
B.S. in Physics, Rank: 1/83

Research Interests

Graph and Geometric Deep Learning, Diffusion and Flow Matching Models, AI for Science

Experience

NEC Laboratories America — Jun 2026 - Aug 2026
Research Intern — *Protein Sequence-Structure Co-Design*

Peking University — Feb 2022 – Sep 2022
Research Intern — *Graph Neural Networks*

Beijing Institute for General Artificial Intelligence — Sept 2021 – Feb 2022
Research Intern — *Molecular Drug Design*

Selected Research

Expressive and Generalizable Spectral Graph Representation Learning

- Developed expressive graph positional encodings that are provably stable under graph perturbations, improving model robustness and out-of-distribution generalization (*ICLR 2024*).
- Proposed multi-potential magnetic Laplacian positional encodings that capture diverse directional structure for directed graphs, improving performance on direction-sensitive tasks such as circuit analysis (*ICLR 2025*).
- Established a novel combinatorial connection between magnetic graph matrix spectra with random walks on directed graphs, enabling efficient spectral methods for motif detection and link prediction (***PNAS** 2026*).

Temporally-aware Flow Matching for Efficient Probabilistic Forecasting

- Developed sequential flow matching inspired by Bayesian filtering, achieving few-step inference while maintaining competitive forecasting and decision-making performance (*ICLR Workshop 2026, **Spotlight***).
- Developed a self-rollout training algorithm for sequential flow matching that mitigates error accumulation over long forecasting horizons (under review).

Geometric Generative Models for Molecular Drug Design

- Built an E(3)-equivariant generative model that jointly produces molecular graphs and 3D geometry, improving structure-aware linker generation for drug discovery (*ICML 2022, **Oral***).

Publications

* indicates equal contribution

[1] Accelerated Sequential Flow Matching: A Bayesian Filtering Perspective
Yinan Huang, Hans Hao-Hsun Hsu, Junran Wang, Bo Dai, Pan Li
*ICLR Workshop ReALM–GEN, 2026 (**Spotlight**)*

[2] What Can We Learn from State Space Models for Machine Learning on Graphs?
Yinan Huang*, Siqi Miao*, Pan Li
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026

[3] Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications
Yinan Huang, David F Gleich, Pan Li
*The Proceedings of the National Academy of Sciences (**PNAS**), 2026*

[4] GenAI Copyright Evidence with Operational Meaning
Eli Chien, Amit Saha, Yinan Huang, Pan Li
ICML Workshop on Reliable and Responsible Foundation Models, 2025

[5] Differentially Private Relational Learning with Entity-level Privacy Guarantees
Yinan Huang*, Haoteng Yin*, Eli Chien, Rongzhe Wei, Pan Li
Advances in Neural Information Processing Systems (NeurIPS), 2025.

[6] What Are Good Positional Encodings for Directed Graphs?
Yinan Huang, Haoyu Wang, Pan Li
International Conference on Learning Representations (ICLR), 2025.

[7] On the Stability of Expressive Positional Encodings for Graphs
Yinan Huang*, William Lu*, Joshua Robinson, Yu Yang, Muhan Zhang, Stefanie Jegelka, Pan Li
International Conference on Learning Representations (ICLR), 2024.

[8] Is Distance Matrix Enough for Geometric Deep Learning?
Zian Li, Xiyuan Wang, Yinan Huang, Muhan Zhang
Advances in Neural Information Processing Systems (NeurIPS), 2023.

[9] Boosting the Cycle Counting Power of Graph Neural Networks with I^2-GNNs
Yinan Huang, Xingang Peng, Jianzhu Ma, Muhan Zhang
International Conference on Learning Representations (ICLR), 2023.

[10] 3DLinker: An E(3) Equivariant Variational Autoencoder for Molecular Linker Design
Yinan Huang, Xingang Peng, Jianzhu Ma, Muhan Zhang
*International Conference on Machine Learning (ICML), 2022 (**Oral**)*.

Honors and Awards

- Travel Award for ICLR 2025
- Georgia Tech ECE Fellowship 2023
- China National Scholarship (top 1%) 2017

Professional Service

- Reviewer for International Conference on Machine Learning (ICML) 2023-2026
- Reviewer for International Conference on Learning Representations (ICLR) 2024-2027
- Reviewer for Advances in Neural Information Processing Systems (NeurIPS) 2023-2026
- Program Committee for Association for the Advancement of Artificial Intelligence (AAAI) 2026
- Reviewer for Association for Computing Machinery's Special Interest Group on Knowledge Discovery and Data Mining (KDD) 2026
- Teaching Assistant: ECE 3077 Introduction to Probability and Statistics, ECE 6250 Advanced Digital Signal Processing

Skills

- Programming languages and frameworks: Python, Pytorch, Pytorch Geometric, Matlab, C